

CSE 4571 – PRACTICAL ROBOTICS PROJECTS WITH ARDUINO

Assignment – 1 (Experiments 1 to 5)

Experiment 1 - Ultrasonic Distance Sensing

TRIG on D9, ECHO on D10; zones at 0–10 cm (Danger), 10–25 cm (Warning), above 25 cm (Safe).

1. Derive the relation $\text{Distance (cm)} = \text{Echo time } (\mu\text{s}) / 58$ from the speed of sound in air.
2. The zone boundaries are fixed at 10 cm and 25 cm. Give one practical reason you might move either boundary on a real moving robot.
3. Why does the HC-SR04 need separate TRIG and ECHO pins rather than one shared pin?
4. Write a function `getDistanceCM()` that triggers the sensor on D9 and times the echo on D10 with `pulseIn()`, returning distance using the formula from Q1.
5. Rewrite the zone if-else block so the 10 cm and 25 cm cut-offs come from two named variables (`dangerLimit`, `warningLimit`) instead of being hard-coded.
6. Given one actual and one measured distance reading, write the line(s) of code that compute and print the percentage error between them.

Experiment 2 - Analog Acquisition with a Photoresistor

Photoresistor AO on A0, indicator LED on D2, threshold at $\text{ADC} = 500$.

7. A 10-bit ADC reports 0–1023 for a 0–5 V input. Show the arithmetic that converts a raw `analogRead(A0)` value into an approximate voltage.
8. Why is the photoresistor read with `analogRead()` while the HC-SR04 is read with `pulseIn()`? State the difference in the sensors' outputs that causes this.
9. Why must a photoresistor's readings be calibrated before they are used for an automatic lighting decision?
10. Write code that reads A0, converts the reading to voltage as in Q7, and prints both values to the Serial Monitor.
11. Write code that averages five consecutive `analogRead(A0)` samples and uses that average, instead of a single sample, to drive the LED threshold.
12. Add one extra ADC band to the existing threshold logic so the program prints DARK, DIM or BRIGHT instead of just ON/OFF.

Experiment 3 - Servo Position Control

Potentiometer wiper on A0, servo signal on D9, 0–1023 mapped to 0–180°.

13. Explain how a servo uses pulse width (PWM) to set its shaft angle, in contrast to a DC motor responding to voltage.
14. Using `map(value, 0, 1023, 0, 180)`, what servo angle does a raw potentiometer reading of 512 correspond to? Show the calculation.
15. Is Experiment 3's potentiometer-to-servo control open-loop or closed-loop? Justify your answer in one or two lines.

16. Write code that attaches a Servo object to D9, reads A0, maps the reading to 0–180° and sends it to the servo.
17. Add a single constrain() call to the Q16 code so the servo's output angle is limited to 20°–160°.
18. Write the code for a pushbutton press (on a digital pin) that moves the servo to the next of three fixed positions - 0°, 90°, 180° - each time it is pressed.

Experiment 4 - Stepper Motor Control via A4988

STEP on D3, DIR on D4, ENABLE on D5; the demo runs 100 steps, reverses, and repeats.

19. Explain what a single pulse on STEP does and what role the DIR pin plays in the A4988's control scheme.
20. Assuming a step angle of 1.8° per step, calculate how many steps are needed for exactly a 90° rotation.
21. Give one point of comparison between servo position control (Experiment 3) and stepper position control (Experiment 4) - for example, precision or the need for feedback.
22. Write code that sets DIR for clockwise rotation and generates 200 STEP pulses on D3 with appropriate timing.
23. Change Experiment 4's step count from the arbitrary 100 to the value calculated in Q20, so the motor turns exactly 90°.
24. Write the line(s) of code that would set the STEP-pulse delay from a potentiometer reading on A0, so a higher reading gives a shorter delay (faster stepping).

Experiment 5 - PIR-Triggered Servo and Buzzer

PIR output on D2, servo signal on D9, buzzer on D6; motion drives the servo to 90° and a 1500 Hz tone, no motion returns it to 0° and silence.

25. Identify the sensing, decision-making and actuation elements in Experiment 5.
26. What is event-driven control, and how does Experiment 5's PIR response differ from the continuous threshold checking used in Experiments 1 and 2?
27. Is Experiment 5's PIR output read as a digital or an analog signal? State how you can tell from the pin and the function used to read it.
28. Write the conditional block that reads the PIR pin (D2) and, on motion, drives the servo (D9) to 90° and the buzzer (D6) to a 1500 Hz tone; on no motion, returns the servo to 0° and stops the tone.
29. Modify the Q28 code so the servo holds at 90° for 3 seconds after motion stops, using millis() rather than delay().
30. Experiment 1's danger-zone branch currently only lights an LED and sounds a buzzer. Add one line to that branch that would instead sweep Experiment 3's servo to 90° as an avoidance action.